

STUDENT PERFORMANCE PREDICTION SYSTEM

CodeVedX Internship Project

Submitted By

Name: Aarti Prajapati

Enrollment Number: 21231258

Course: B.Tech CSE (AI & ML), 7th Semester

College: IIMT University, Meerut ,Uttar Pradesh

Email : aartiaiml3009@gmail.com

Project Theme

Artificial Intelligence & Machine Learning (AI/ML)

Sub-Domain: Recommendation System

Project Type: Content-Based Movie Recommendation System

GitHub Repository

https://github.com/aartiprajapati-ai/CODEVEDX/tree/main/Student_Performance_Prediction_System

Academic Year: 2026–2027

INTRODUCTION

Education plays a vital role in the personal and professional development of students. Predicting a student's academic performance at an early stage can help educators identify learning difficulties and provide timely guidance to improve overall results. With the advancement of **Artificial Intelligence (AI)** and **Machine Learning (ML)**, educational institutions can analyze student-related data to make accurate predictions and support data-driven decision-making.

The **Student Performance Prediction System** is a Machine Learning-based application designed to predict a student's academic performance using factors such as **attendance, study hours, and previous marks**. The system analyzes these inputs and generates performance predictions using a trained Machine Learning model. This enables teachers and students to identify potential academic risks and take corrective actions before the final examination.

The project follows a complete Machine Learning workflow, including **data collection, data preprocessing, feature selection, model training, model evaluation, and prediction**. The application is developed using **Python** and various Machine Learning libraries, providing a simple and efficient solution for educational performance analysis.

The primary objective of this project is to demonstrate the practical implementation of Artificial Intelligence and Machine Learning in the education sector by developing a reliable prediction system that supports academic improvement and enhances the learning process.

PROBLEM STATEMENT

In educational institutions, evaluating a student's academic performance is often based on examination results, which may not accurately reflect the student's learning progress throughout the academic session. Many students face difficulties due to factors such as low attendance, insufficient study hours, and poor academic preparation. Identifying these students at an early stage through manual analysis is time-consuming and challenging for teachers.

With the increasing availability of educational data, there is a need for an intelligent system that can analyze student-related information and predict academic performance before the final examinations. Such a system can help educators identify students who may require additional support and enable timely academic interventions.

The **Student Performance Prediction System** is developed to address this challenge by using **Machine Learning** techniques to analyze factors such as **attendance, study hours, and previous marks**. The system predicts the expected academic performance of students, assisting teachers and students in making informed decisions to improve learning outcomes and overall academic success.

PROJECT OBJECTIVE

The main objectives of the **Student Performance Prediction System** are:

- To develop a Machine Learning-based system for predicting student academic performance.
- To analyze important factors such as **attendance, study hours, and previous marks** that influence student performance.
- To implement data preprocessing and feature selection techniques for improving prediction quality.
- To train and evaluate a Machine Learning model for accurate performance prediction.
- To provide early identification of students who may require additional academic support.
- To assist teachers in monitoring student progress and making data-driven decisions.
- To build a simple and user-friendly application for predicting student performance.
- To demonstrate the practical application of Artificial Intelligence and Machine Learning in the field of education.

EXPECTED OUTCOMES

- Accurate prediction of student academic performance.
- Early identification of students at academic risk.
- Improved decision-making for teachers and educational institutions.
- Better understanding of Machine Learning concepts and predictive analytics.
- A reliable system that can be further enhanced with additional student-related features and larger datasets.

SCOPE OF THE PROJECT

The **Student Performance Prediction System** is designed to predict students' academic performance by analyzing key factors such as **attendance, study hours, and previous academic marks**. The system applies Machine Learning techniques to identify performance patterns and generate predictions that can help educators and students make informed academic decisions.

The project aims to support educational institutions by providing an automated solution for monitoring student performance. Teachers can use the prediction results to identify students who may require additional guidance, while students can better understand their academic progress and improve their learning strategies.

The current version of the system focuses on a predefined dataset and predicts student performance based on selected input features. It provides a simple and user-friendly interface for entering student information and viewing prediction results.

In the future, the system can be enhanced by integrating larger educational datasets, additional academic and behavioral factors, cloud-based deployment, real-time database connectivity, and advanced Machine Learning algorithms. These improvements will increase prediction accuracy and make the system suitable for large-scale educational institutions.

Overall, this project demonstrates the practical application of **Artificial Intelligence and Machine Learning** in the education sector, helping improve academic performance through data-driven analysis and predictive modeling.

LIMITATIONS

Although the **Student Performance Prediction System** provides useful predictions, it has certain limitations:

- The prediction accuracy depends on the quality and size of the dataset used for training.
- The system considers only a limited number of factors such as **attendance, study hours, and previous marks**. Other factors like motivation, health, family background, and learning environment are not included.
- The model may not perform well when presented with data that is significantly different from the training dataset.
- The current system is designed for academic prediction only and cannot provide personalized learning recommendations.
- The application does not support real-time database connectivity or automatic data updates.
- The system currently uses a single Machine Learning algorithm, which may not always provide the highest possible prediction accuracy.
- The application is intended for educational purposes and should be used as a decision-support tool rather than the sole basis for evaluating a student's academic performance.

SOFTWARE REQUIREMENTS

The following software tools and technologies were used to design, develop, and implement the **Student Performance Prediction System**.

Software	Purpose
Python 3.x	Programming Language
Visual Studio Code	Code Editor and Development Environment
Jupyter Notebook / Google Colab	Model Development and Testing
Pandas	Data Loading and Preprocessing
NumPy	Numerical Computations
Matplotlib	Data Visualization
Scikit-learn	Machine Learning Model Development
Joblib	Model Saving and Loading
Streamlit	Web Application Development
Git & GitHub	Version Control and Project Hosting
Google Chrome	Running the Web Application

HARDWARE REQUIREMENTS

The following hardware components are required to develop and run the **Student Performance Prediction System** efficiently.

Hardware Component	Specification
Processor	Intel Core i3 or above (Intel Core i5 Recommended)
RAM	Minimum 4 GB (8 GB Recommended)
Storage	Minimum 500 MB Free Disk Space
Operating System	Windows 10 / Windows 11
Display	1366 × 768 Resolution or Higher
Internet Connection	Required for downloading libraries, GitHub access, and project updates

PYTHON LIBRARIES USED

The **Student Performance Prediction System** utilizes several Python libraries for data processing, visualization, machine learning, model persistence, and web application development. These libraries simplify the implementation of the complete machine learning workflow.

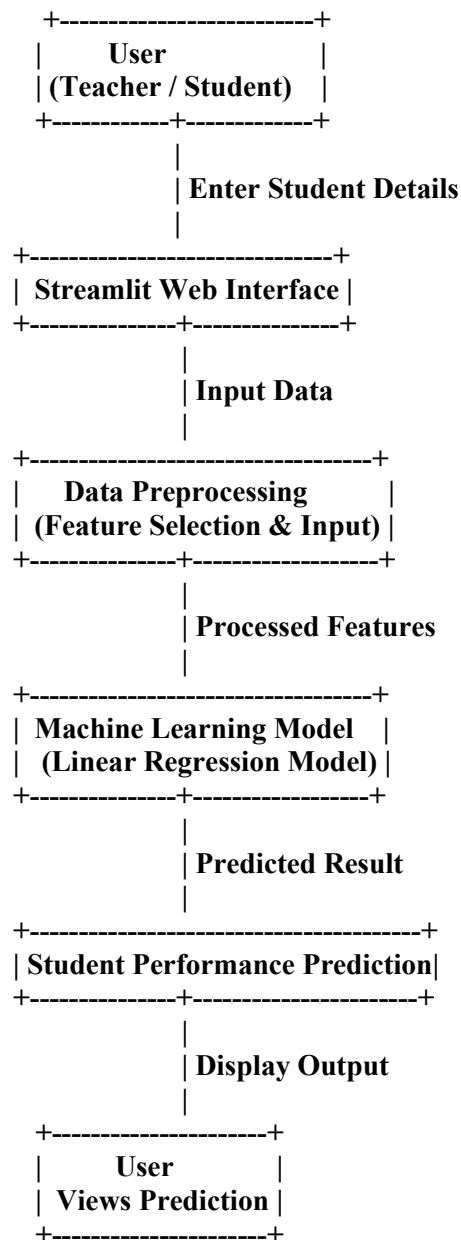
Library	Purpose
Pandas	Used for loading, cleaning, and manipulating the dataset.
NumPy	Used for numerical computations and array operations.
Matplotlib	Used to create graphs and visualize the dataset.
Scikit-learn	Used for data preprocessing, model training, and performance evaluation.
Joblib	Used to save and load the trained Machine Learning model.
Streamlit	Used to develop the interactive web application interface.

SYSTEM ARCHITECTURE

The **Student Performance Prediction System** follows a simple and efficient architecture for predicting students' academic performance. The user enters student details such as **attendance**, **study hours**, and **previous marks** through the Streamlit web interface. These inputs are processed and passed to the trained Machine Learning model. The model analyzes the input values and predicts the student's expected academic performance. Finally, the prediction result is displayed to the user through the application interface.

The architecture ensures fast prediction, ease of use, and efficient implementation of Machine Learning techniques for educational data analysis.

System Architecture Flow



WORKING OF THE SYSTEM

The **Student Performance Prediction System** works through the following steps:

1. The user enters student information such as **attendance, study hours, and previous marks** through the Streamlit interface.
2. The application receives the input and performs basic validation.
3. The processed input data is sent to the trained Machine Learning model.
4. The Linear Regression model analyzes the provided features.
5. The model predicts the student's expected academic performance.
6. The predicted result is displayed on the Streamlit application.
7. Users can use the prediction to understand academic progress and identify areas for improvement.

DATASET DESCRIPTION

The **Student Performance Prediction System** uses a structured dataset containing information related to students' academic activities and performance. The dataset is stored in **CSV (Comma-Separated Values)** format and is used to train the Machine Learning model.

The dataset contains important academic attributes such as **attendance percentage, study hours, previous marks, and final performance**. These features help the model identify patterns and predict a student's expected academic performance.

Before training the model, the dataset undergoes preprocessing, including data cleaning, handling missing values (if any), and feature selection. The processed dataset is then used to train the Machine Learning model for accurate predictions.

The quality and diversity of the dataset play a significant role in improving the prediction accuracy of the system.

Dataset Information

Attribute	Description
Dataset Name	Student Performance Dataset
File Name	student_performance.csv
File Format	CSV (Comma-Separated Values)
Dataset Type	Structured Tabular Dataset
Domain	Education
Purpose	Student Performance Prediction

Dataset Schema

Column Name	Data Type	Description
Attendance	Integer	Student attendance percentage (%)
Study_Hours	Integer	Average study hours per day
Previous_Marks	Integer	Marks obtained in previous examination
Performance	Integer/Float	Predicted or actual student performance

Sample Dataset

Attendance (%)	Study Hours	Previous Marks	Performance
90	6	85	88
75	4	70	72
95	8	92	94
60	3	58	60
85	5	80	82

DATA PREPROCESSING

Before training the Machine Learning model, the following preprocessing steps were performed:

- Loaded the dataset using the **Pandas** library.
- Checked for missing or inconsistent values.
- Selected relevant features such as attendance, study hours, and previous marks.
- Separated the input features (X) and target variable (Y).
- Split the dataset into **training** and **testing** sets.
- Trained the Machine Learning model using the processed data.
- Saved the trained model using the **Joblib** library for future predictions.

METHODOLOGY

The **Student Performance Prediction System** was developed using a systematic Machine Learning approach. The complete methodology includes data collection, preprocessing, model training, evaluation, and prediction. The workflow followed during the development of the project is described below.

Step 1: Dataset Collection

A student performance dataset was collected in **CSV format**. The dataset contains important academic attributes such as attendance, study hours, previous marks, and student performance, which are used for prediction.

Step 2: Data Loading

The dataset was loaded into Python using the **Pandas** library for analysis and preprocessing.

```
import pandas as pd
```

```
df = pd.read_csv("student_performance.csv")
```

Step 3: Data Preprocessing

The dataset was cleaned and prepared before training the Machine Learning model.

The preprocessing steps included:

- Loading the dataset
- Checking for missing values
- Selecting important features
- Separating input and output variables
- Preparing the dataset for training

Step 4: Feature Selection

The following input features were selected:

- Attendance
- Study Hours
- Previous Marks

These features were used to predict the student's overall academic performance.

Step 5: Train-Test Split

The dataset was divided into **training** and **testing** datasets.

- **Training Data:** 80%
- **Testing Data:** 20%

This helps evaluate the model's performance on unseen data.

Step 6: Model Training

A **Linear Regression** model was trained using the selected features.

```
from sklearn.linear_model import LinearRegression  
  
model = LinearRegression()  
  
model.fit(X_train, y_train)
```

Step 7: Model Evaluation

After training, the model was evaluated using the testing dataset to measure its prediction performance.

The predicted values were compared with the actual values to verify the accuracy and reliability of the model.

Step 8: Model Saving

The trained Machine Learning model was saved using the **Joblib** library.

```
import joblib  
  
joblib.dump(model, "student_performance_model.pkl")
```

Saving the model eliminates the need to retrain it every time the application runs.

Step 9: Prediction

The saved model is loaded into the application. Users enter student details such as attendance, study hours, and previous marks. The model processes the input and predicts the student's expected academic performance.

PROJECT IMPLEMENTATION

The **Student Performance Prediction System** was implemented using **Python**, **Machine Learning**, and the **Streamlit** framework. The project follows a modular structure, where each file is responsible for a specific task such as data processing, model loading, prediction, and user interaction.

The implementation is divided into the following modules:

1. Dataset Module (student_performance.csv)

The dataset contains student academic information used for training the Machine Learning model.

Responsibilities:

- Store student academic records

- Provide input data for model training
- Maintain structured CSV format

2. Model Module (student_performance_model.pkl)

The trained **Linear Regression** model is stored in a .pkl file using the Joblib library.

Responsibilities:

- Store the trained Machine Learning model
- Load the model without retraining
- Generate performance predictions

3. Application Module (app.py)

The app.py file is responsible for running the Streamlit web application.

Responsibilities:

- Load the trained model
- Accept user input
- Process input values
- Predict student performance
- Display the prediction result

4. User Interface (Streamlit)

The Streamlit interface allows users to enter student information through interactive input fields.

The interface includes:

- Attendance Input
- Study Hours Input
- Previous Marks Input
- Predict Button
- Performance Prediction Output

5. Prediction Process

The prediction process follows these steps:

1. User enters student details.
2. Streamlit receives the input values.
3. The trained model processes the input.
4. The Linear Regression algorithm predicts the student's performance.
5. The predicted result is displayed on the screen.

RESULTS AND OUTPUT

The **Student Performance Prediction System** was successfully developed and tested using a Machine Learning model. The application predicts a student's academic performance based on input parameters such as **attendance, study hours, and previous marks**.

The trained **Linear Regression** model generated accurate predictions for the given inputs. The Streamlit application provides a simple and interactive interface where users can enter student details and instantly receive the predicted performance.

The successful implementation of this project demonstrates the practical application of **Machine Learning** in the education sector, helping teachers and students analyze academic performance more effectively.

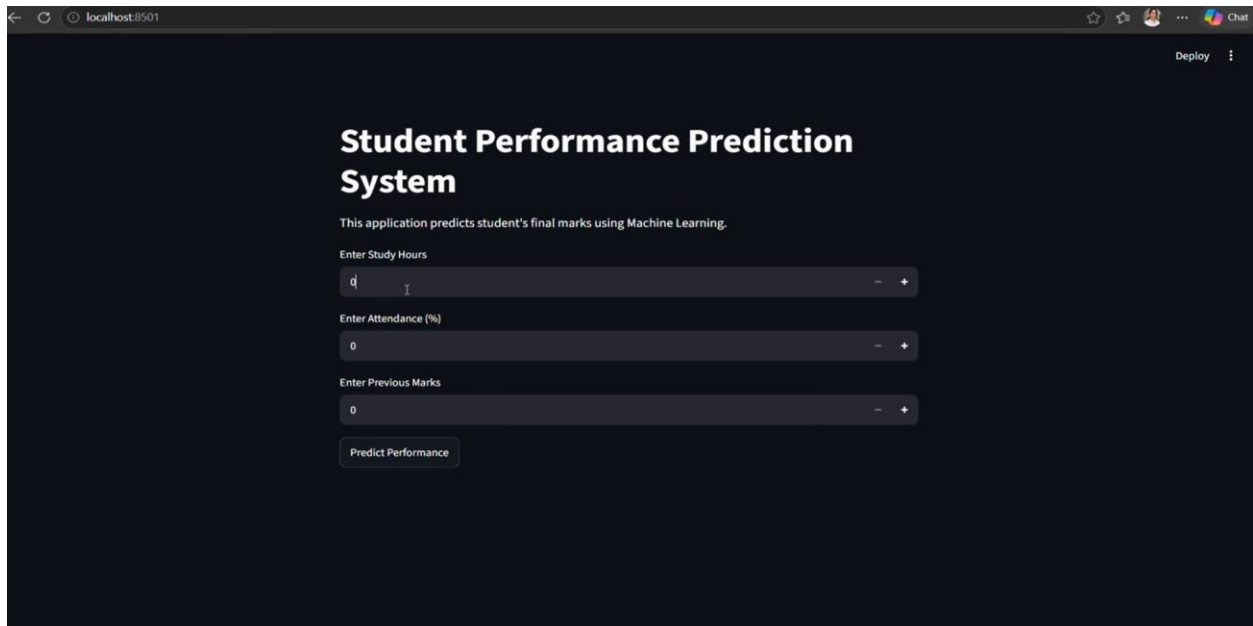
Output Screenshots

1. Home Page

Description:

The home page of the Streamlit application allows users to enter student details such as attendance, study hours, and previous marks.

Screenshot:



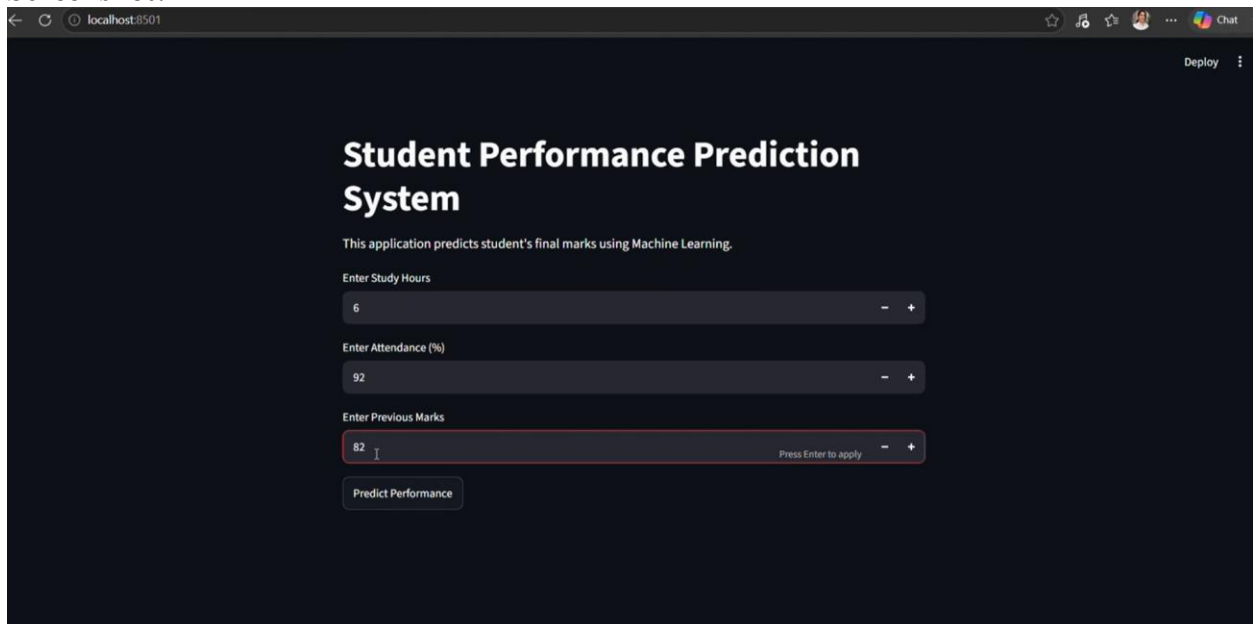
A screenshot of a web browser displaying the 'Student Performance Prediction System' home page. The page has a dark blue background. At the top, the title 'Student Performance Prediction System' is displayed in white. Below the title, a subtitle reads 'This application predicts student's final marks using Machine Learning.' There are three input fields: 'Enter Study Hours' with a value of '4', 'Enter Attendance (%)' with a value of '0', and 'Enter Previous Marks' with a value of '0'. Each input field has a minus sign on the left and a plus sign on the right. Below the input fields is a button labeled 'Predict Performance'. The browser's address bar shows 'localhost:8501'.

2. Student Details Input

Description:

The user enters the required academic details through the input fields provided in the application.

Screenshot:



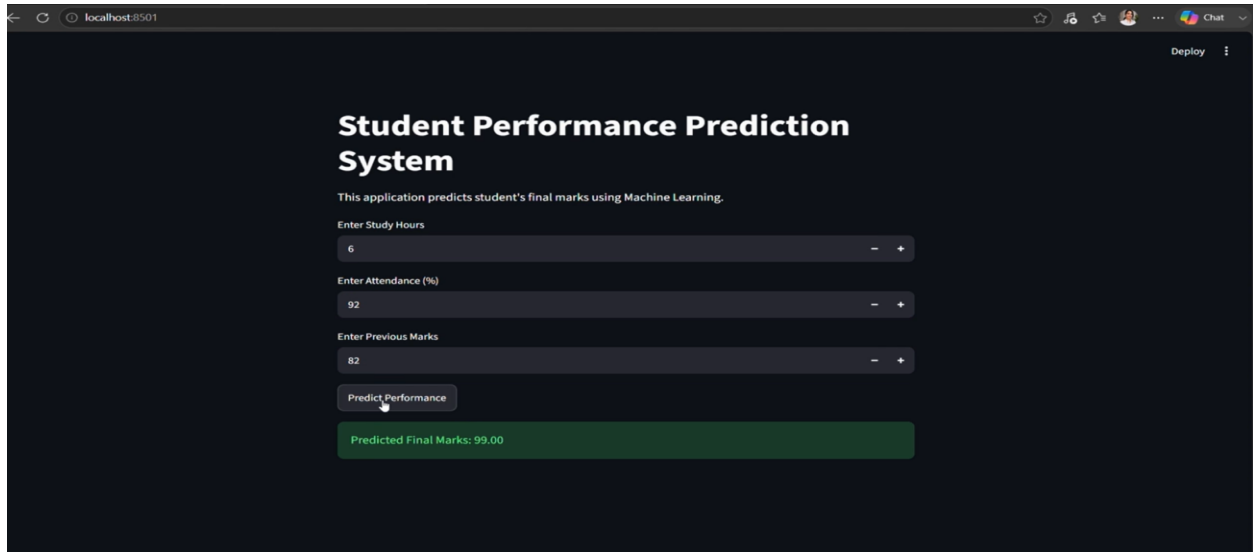
A screenshot of the same web browser displaying the 'Student Performance Prediction System' home page. The input fields now contain the following values: 'Enter Study Hours' is '6', 'Enter Attendance (%)' is '92', and 'Enter Previous Marks' is '82'. The 'Enter Previous Marks' field has a red border and a small text prompt 'Press Enter to apply' on the right. The 'Predict Performance' button remains at the bottom. The browser's address bar shows 'localhost:8501'.

3. Prediction Result

Description:

After clicking the **Predict Performance** button, the Machine Learning model analyzes the input values and displays the predicted student performance.

Screenshot:



4. Model Loading

Description:

The trained Machine Learning model (student_performance_model.pkl) is loaded successfully using the Joblib library.

Screenshot:

6. Model Training

We use Linear Regression algorithm to predict student final marks.

```
In [11]: # Creating and training the model
         model = LinearRegression()
         model.fit(X_train, y_train)
         print("Model trained successfully")
```

Model trained successfully

7. Model Prediction

The trained model predicts final marks for test data.

```
In [12]: # Making predictions
         y_pred = model.predict(X_test)
         y_pred
```

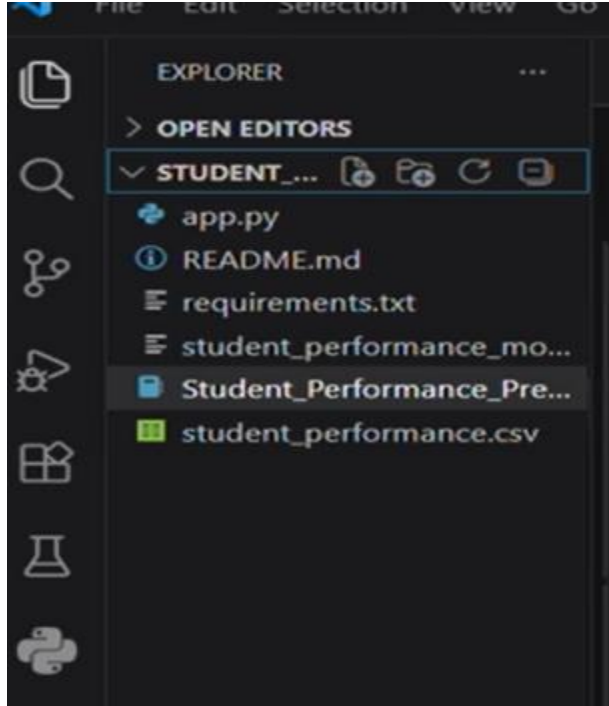
```
Out[12]: array([40., 60.])
```

5. Project Structure

Description:

The project follows a well-organized folder structure, making it easy to understand, maintain, and extend.

Screenshot:

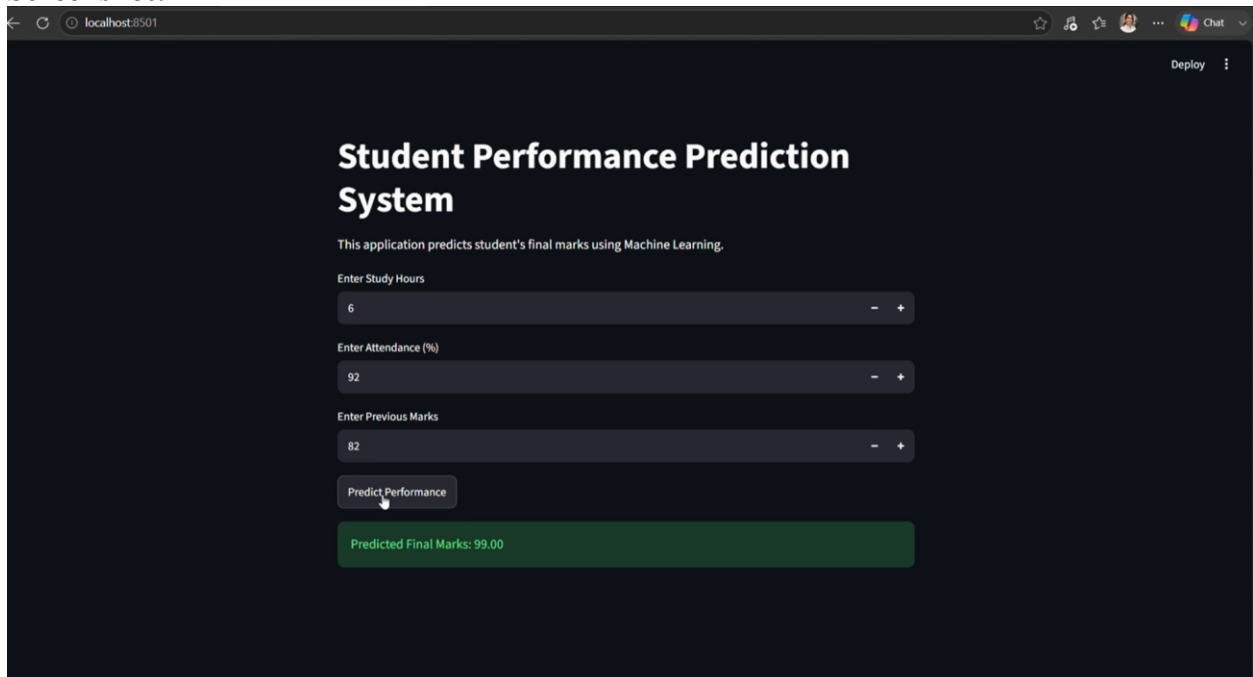


6. Streamlit Application Running

Description:

The Streamlit server runs successfully, allowing users to access the application through a web browser and perform real-time predictions.

Screenshot:



ADVANTAGES

- Predicts student performance quickly and accurately.
- Reduces manual analysis by teachers.
- Provides an easy-to-use web interface.
- Helps identify students who may need additional academic support.
- Demonstrates the practical application of Machine Learning in education.
- Easy to update with new datasets and additional features.

CONCLUSION

The **Student Performance Prediction System** was successfully developed using **Machine Learning** techniques to predict students' academic performance based on factors such as attendance, study hours, and previous marks.

The project demonstrates the complete Machine Learning workflow, including data preprocessing, feature selection, model training, evaluation, and deployment through a Streamlit-based web application.

The system provides quick and reliable predictions, helping educators identify students who may require additional academic support. It also enables students to understand their academic progress and take corrective measures to improve their performance.

Overall, the project demonstrates the practical application of Artificial Intelligence and Machine Learning in the education sector and serves as a useful decision-support tool for academic performance analysis.

FUTURE SCOPE

The current system can be further enhanced by implementing the following features:

- Integrate larger and more diverse educational datasets to improve prediction accuracy.
- Include additional factors such as assignment scores, extracurricular activities, classroom participation, and behavioral analysis.
- Implement advanced Machine Learning and Deep Learning algorithms for better prediction performance.
- Develop a personalized recommendation system to provide study suggestions based on predicted performance.
- Integrate the application with school or college database management systems.
- Deploy the application on cloud platforms such as AWS, Microsoft Azure, or Google Cloud.
- Develop a mobile application for easy accessibility by students and teachers.
- Add graphical dashboards and real-time analytics for better visualization of student performance.

REFERENCES

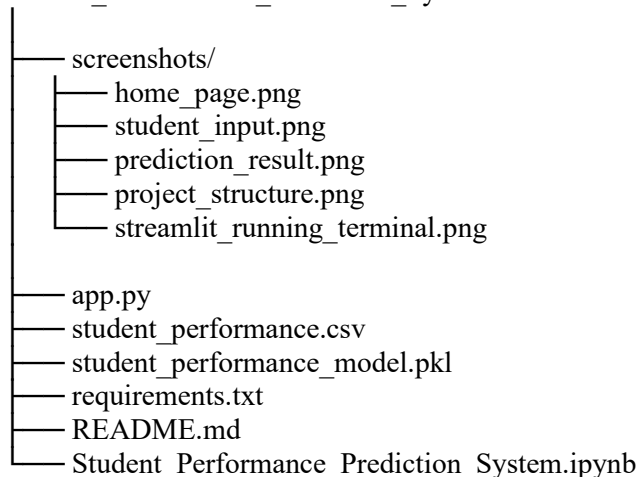
The following resources were referred to during the development of this project:

1. Python Software Foundation. *Python Documentation*. <https://docs.python.org/>
2. Scikit-learn Developers. *Scikit-learn Documentation*. <https://scikit-learn.org/>
3. Pandas Development Team. *Pandas Documentation*. <https://pandas.pydata.org/>
4. NumPy Developers. *NumPy Documentation*. <https://numpy.org/>
5. Matplotlib Development Team. *Matplotlib Documentation*. <https://matplotlib.org/>
6. Streamlit Documentation. <https://docs.streamlit.io/>
7. Joblib Documentation. <https://joblib.readthedocs.io/>

APPENDIX

Appendix A – Project Folder Structure

Student_Performance_Prediction_System



Appendix B: Project Modules

The project consists of the following modules:

- Data Collection
- Data Preprocessing
- Feature Selection
- Train-Test Split
- Linear Regression Model Training
- Model Evaluation
- Model Saving using Joblib
- Streamlit Web Application
- Student Performance Prediction

Appendix C: Screenshots

The following screenshots are included in the project documentation:

1. Home Page
2. Student Details Input
3. Prediction Result
4. Project Folder Structure
5. Streamlit Application Running
6. Model File (student_performance_model.pkl)